Pitch Deck: Transforming Industrial Emissions into Structural Asset Classes

Company Name: [CarbonLattice Technologies / Your Startup Name]

Tagline: Coral-Inspired Multi-Pollutant Solidification for the Industrial Circular Economy

Target Audience: Climate-Tech VCs, Strategic Industrial Partners, and ESG Infrastructure Funds

Slide 1: The Title & The Hook (The "Coral-in-a-Box" Concept)

Visual:

A split screen showing a dying marine coral reef on the left, and a modern, high-precision industrial concrete substitute block on the right, connected by a glowing molecular double helix.

Presenter Narrative:

> "Every second, heavy industry pours thousands of tons of volatile carbon, sulfur, and heavy metals into our atmosphere. For decades, the climate-tech sector has viewed this as a storage crisis—scrambling to find deep geological holes to inject gaseous waste.
>
> But nature solved this problem millions of years ago. Marine corals don't inject carbon underground; they use enzymes and biopolymer templates to transform dissolved carbon dioxide into rock-hard structural skeletons at room temperature.
>
> Today, we are introducing a technology that packages this exact biological pathway into a bolt-on mechanical wet-mesh scrubber. We don't store emissions. We solidify them into high-density, non-leachable structural eco-blocks. We are turning a multi-billion dollar environmental liability into a localized physical asset class."

Slide 2: The Problem (The Industrial Trilemma)

Visual:

An infographic illustrating three distinct bottlenecks choking Indian manufacturing:

1. **The Energy Penalty:** Traditional Carbon Capture (CCS) consumes up to 30% of a plant's output just to heat chemical solvents.
2. **The High-Ash Barrier:** Indian coal has an exceptionally high ash content (35–45%). Traditional scrubbers clog and fail on this particulate matter.
3. **The Geological Storage Bottleneck:** India has highly constrained, unproven underground storage basins, leaving captured gas with nowhere to go.

Presenter Narrative:

> "Industrial emitters—especially in emerging economies like India—face a brutal trilemma. If they install traditional carbon capture, the energy penalty ruins their

margins. If they run standard wet scrubbers, India's high-ash coal clogs the machinery, creating toxic liquid sludge that ruins local groundwater. And even if they capture the carbon, they have no geological formations to store it.

Environmental compliance is no longer a corporate social responsibility goal; under the new Central Pollution Control Board (CPCB) guidelines and the Indian Carbon Credit Trading Scheme (CCTS), missing emissions targets now carries direct, daily financial penalties."

Slide 3: The Solution (The Carbon-to-Lattice System)

Visual:

A clean engineering schematic of our 4-stage modular plant: `Raw Gas Inlet` ——> `Stage 1: Quench` ——> `Stage 3: Static Mesh Tower` ——> `Stage 4: Block Foundry`

Presenter Narrative:

> "Our solution is a modular, multi-phase wet-mesh absorption system that retrofits directly onto existing industrial smokestacks.
>
> Instead of using high-temperature chemical boilers or complex electrocatalytic cells, our system draws gas through a series of static wire mesh screens coated with an organic biopolymer fluid.
>
> The gas is hydrated instantly by an immobilized enzymatic engine, prompting a rapid **Sol-to-Gel phase transition**. Heavy metals are bound at the molecular level, sulfur is co-precipitated as gypsum, and the resulting dense crystalline slurry is continuously compressed into solid, high-strength eco-blocks. Zero underground injection required."

Slide 4: The Technology & Biochemical Moat

Visual:

A high-resolution molecular 3D model showing:

1. Gaseous CO_2 entering the zinc-bound active site of **Carbonic Anhydrase (CA)** and hydrating at microsecond speed.
2. Divalent calcium ions (Ca^{2+}) binding to the dense amine ($R - NH_2$) and hydroxyl clusters of a **protonated chitosan backbone**, forcing rapid crystallization of $CaCO_3$ along the polymer chain.

Presenter Narrative:

> "Our technological moat lies in combining nature's fastest biological catalyst with low-cost organic polymers.
>
> First, our immobilized, thermo-alkaliphilic Carbonic Anhydrase enzyme variant, *CA-KR1*, accelerates the kinetic hydration of carbon dioxide by up to six orders of magnitude.

Second, we introduce protonated chitosan derived from coastal shrimp shell waste. The repeating positive charge profile along the chitosan chain reduces the thermodynamic nucleation energy barrier, forcing calcium carbonate to crystallize rapidly directly along the polymer backbone. As the crystals lock together, they form a dense, reinforced composite."

Slide 5: Strategic Market Positioning (Our Competitive Moat)

Visual:

A clear, competitive matrix comparing your startup directly with your core regional and global counterparts:

Attribute	Traditional Scrubbers	Svante (Global MOFs)	UrjanovaC (IIT Bombay)	Our System
Core Chemistry	Chemical Absorption	Metal-Organic Frameworks	Electrocatalysis	**Biomimetic Polymer Loop**
Energy Input	High (Thermal regeneration)	Very High (Steam heat)	High (Electrochemical cell)	**Ambient (Low mechanical)**
Ash Sensitivity	Highly Sensitive	Extremely Sensitive	Moderate (Aqueous)	**Beneficial (Used as aggregate)**
Storage Need	Solid Waste Sludge	Underground Injection	Deep Geological Injection	**None (Sold as Eco-Blocks)**
CAPEX Profile	Moderate	Ultra-High (Exotic materials)	High (Electrocatalytic cells)	**Low (Simple mechanical retrofit)**

Presenter Narrative:

"Let's be clear about where we stand in the market. Global giants like Svante use synthetic MOF filters that are highly sensitive to moisture and ash, requiring expensive upstream gas conditioning.

Our closest regional counterpart, UrjanovaC, has built an elegant electrocatalytic system, but it requires a constant input of renewable electricity to drive its electrochemical cells, and its primary carbonate pathways remain dependent on geological basalt storage.

Our system treats moisture, fly ash, and sulfur as functional components of the solidification reaction. We run at ambient temperature, use low-cost coastal biowaste, and produce a high-strength, non-leachable construction product. We win on CAPEX, simplicity, and immediate monetization."

Slide 6: Indian Market Viability & Raw Material Access

Visual:

A map of India highlighting:

1. **The Western & Southern Coasts (Gujarat, Maharashtra, Tamil Nadu):** Thousands of tons of coastal shrimp shell waste, guaranteeing a cheap, domestic supply of chitosan.
2. **The Industrial Belt (Odisha, Jharkhand, Chhattisgarh):** Million-ton piles of steel slag (calcium source) and high-ash coal-fired power stations.

Presenter Narrative:

> "India represents the absolute ideal market for this technology due to three localized factors.
>
> First, Indian coal is notoriously high-ash. Rather than viewing fly ash as a pollutant we must filter out, our mass balance utilizes this ash as a pozzolanic aggregate filler, increasing the density of our blocks and reducing raw polymer costs by 22%.
>
> Second, we have virtually free access to raw inputs. We source our reactive calcium ions from abundant steel-mill slag, and we source our chitosan directly from coastal seafood processing waste.
>
> Third, we map perfectly to India's regulatory push, keeping facilities safely below the SPCB limits while generating high-value compliance certificates for the newly active Indian Carbon Credit Trading Scheme (CCTS)."

Slide 7: Unit Economics & Payback (The 10,000 $\mathrm{Nm}^3/\mathrm{hr}$ Plant)

Visual:

A clean, high-impact cash-flow waterfall chart displaying:

- **Initial CAPEX:** INR 1.60 Crore (~$190,000 USD)
- **Hourly Operating Costs (OPEX):** INR 4,556 (Raw chitosan flakes + processed slag + power)
- **Hourly Revenue Streams:** INR 11,562 (Eco-block sales @ ₹5.50/block + CCTS credits)
- **Net Hourly Margin:** INR 7,006
- **Payback Window: Less than 4 months** of continuous operation.

Presenter Narrative:

> "Our unit economics prove that environmental compliance does not have to be a pure cost center. For a mid-size slipstream treating $10,000\ \mathrm{Nm}^3/\mathrm{hr}$ of flue gas, we can deliver a complete, automated installation for a CAPEX of just INR 1.6 Crore.
>
> On the operating side, because we utilize low-cost industrial byproducts and run at ambient temperatures, our hourly costs are capped at ₹4,556.

> Meanwhile, the plant generates over 2,000 high-density eco-blocks and earns valuable CCTS carbon credits every hour, generating ₹11,562 in revenue. This results in a net hourly margin of ₹7,006. With standard annual operational hours, the plant offsets its initial CAPEX in less than four months."

Slide 8: The Ask & Milestones

Visual:

A 6-month progress timeline leading to a validated, commercial pilot launch:

- **Month 1–2:** Digital Twin Optimization & Core Patent Filing (Carreau-Yasuda non-Newtonian flow modeling)
- **Month 3–4:** Bench-Scale 1:100 Lab Validation (TCLP leach testing to prove the "leach-proof" claim)
- **Month 5–6:** Joint pilot development with an industrial partner, backed by IIT Bombay's National CoE in CCUS.

Presenter Narrative:

> "We are raising [Insert Funding Ask, e.g., $250,000 USD / INR 2 Crore] to execute our initial 6-month roadmap.
>
> This capital will lock down our core chemical-mechanical patents, complete our 1:100 scale lab-bench prototype, and secure third-party TCLP leach-testing verification from accredited agencies.
>
> By partnering with IIT Bombay's National Center of Excellence, we will validate our digital twin before deploying our first commercial pilot with a mid-size industrial partner in Month 6.
>
> Join us in shifting heavy industry from a model of atmospheric storage to a circular economy of physical mineralization. Thank you."